

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

*in the name of Allah, the most beneficent,
the most merciful*



Vitamins: Dietary sources, biochemical functions and deficiency diseases



By

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By the end of this lecture, the student will be able to:

- a1. Define vitamins and enumerate their characters.
- a2 . List the differences between water and fat soluble vitamins.
- a3. List dietary sources of each vitamin.
- a4. Point out functions and metabolic role of each vitamin.
- a5. Describe sources deficiency manifestations of each vitamin.



Definition: Vitamins are organic nutrients which are required in small quantities for normal growth and health (Micronutrients).

Characters of vitamins:

- Synthesized in body in small amounts
- Must be supplied in diet
- Don't enter in the structure of tissues
- Don't undergo oxidation to give energy

Classification: Vitamins are classified into 2 groups:

	Fat soluble vitamins	Water soluble vitamins
1. Members	Vitamin: A, D, E,K	Vitamin: C & B complex
2. Source	Dietary fat	Meat, Green leafy vegetables and citrus fruits
3. Absorption	Needs bile salts	need no bile salts
4. Carrier protein	Need carrier	Need no carrier
5. Storage	Stored in liver	Not stored
6. Excretion in urine	Can't be excreted	Easily excreted
7. Deficiency	Appear later	Appear early

Vitamin D

- Vitamin D2 (Ergocalciferol)
- Vitamin D3 (Cholecalciferol)

The active form of vit D
is -1,25 dihydroxycholecalciferol (Calcitriol).

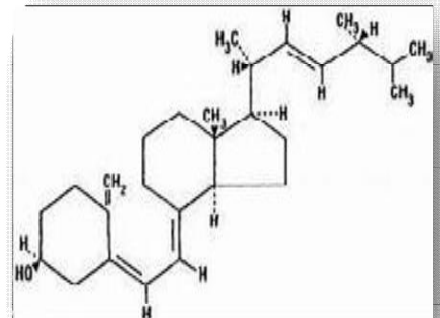
Sources: Tuna, Shark, Cod liver oil

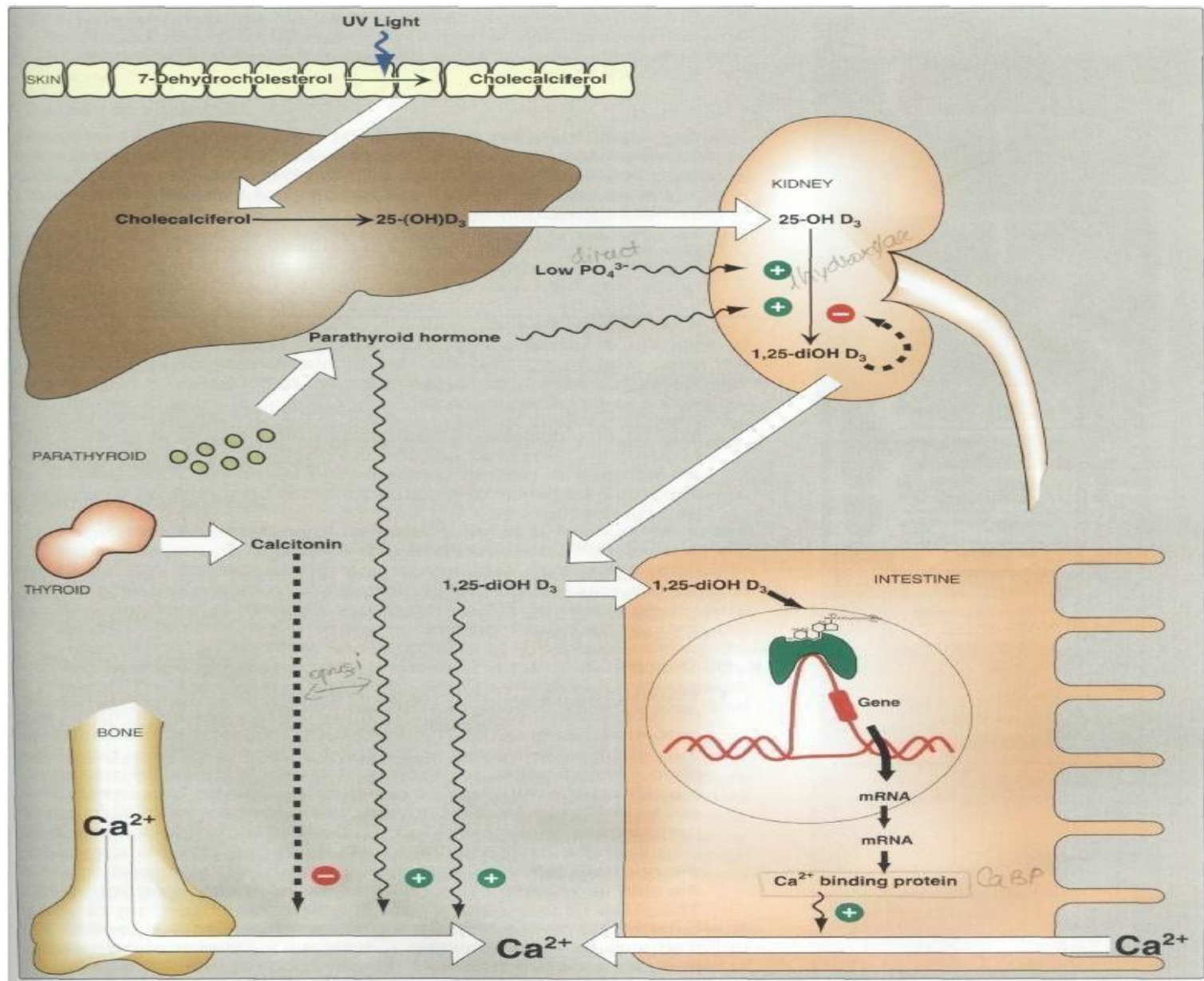
Activation

1 In skin: 7-Dehydrocholesterol is converted to cholecalciferol (using UV of the sun), then is absorbed into the blood stream.

2 In the liver: cholecalciferol is firstly hydroxylated by 25 hydroxylase enzyme to (25OH) cholecalciferol then,

3 In the Kidney: It is more hydroxylated by 1α hydroxylase enzyme to give the active form 1, 25-(OH)₂ cholecalciferol (Calcitriol)





Function of vit. D:

Is to ↑ Plasma Ca^{+2} by acting on:

- ✓ **Intestine**: It ↑ intestinal absorption of Ca^{+2}
- ✓ **Kidney**: It ↑ Ca^{+2} reabsorption from renal tubules
- ✓ **Bone**: It stimulates deposition of Ca^{+2} in bone and teeth (help bone mineralization).

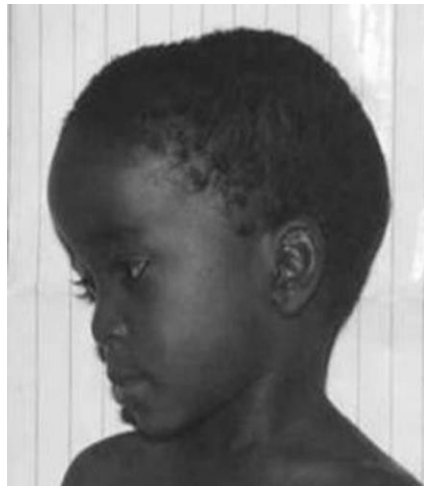
Deficiency manifestations: These include:

A- Rickets in children:

- 1 Low plasma Ca^{+2} .
- 2 Skeletal deformities.
- 3 $\uparrow$ ALP enzyme



Bow-legs



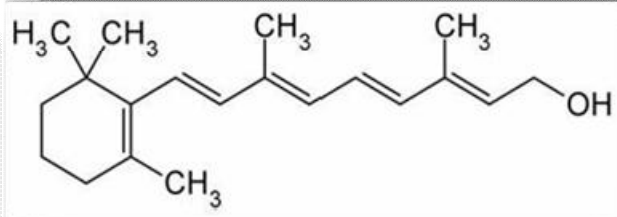
Box shaped skull



Knock-knee

B- Osteomalacia in adults: Multiple fractures due to softening of bone.

Vitamin A



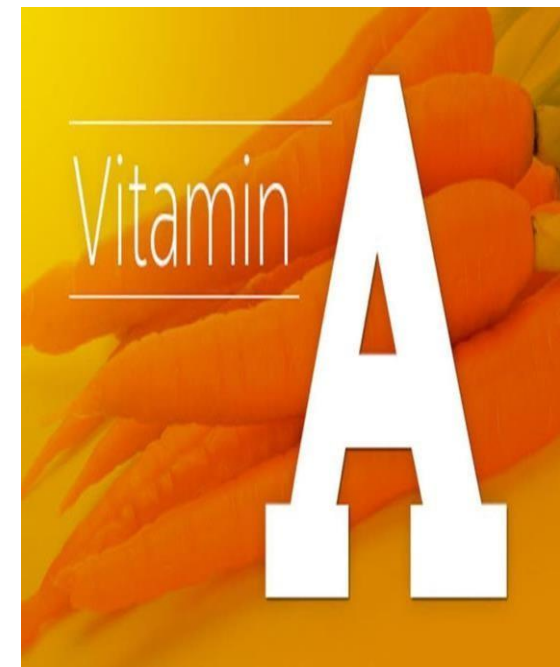
Sources:

1 Provitamins (Carotenes):

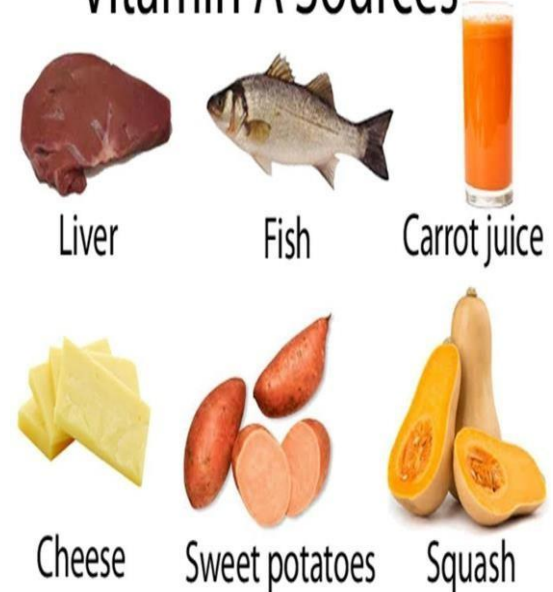
- ✓ **Plant:** carrots sweet potatoes & tomatoes
- ✓ **Animal:** Milk and butter

2-Vitamin A:

**Present in animal source only
as kidney, egg yolk, liver, tuna
shark and cod liver oil**



Vitamin A Sources



❧ Functions:

- ✓ Vitamin A has a role in Growth & Reproduction.
- ✓ **Retinol** is essential for normal growth and bone & teeth formation, normal skin.
- ✓ **Retinal**: Has an important role in vision during night as it is a component of a visual pigment present in cones and rods called rhodopsin.
- ✓ Antioxidant (anticancer) action.

Deficiency manifestations:

- ✓ **Eye manifestations:** Night blindness occurs when liver stores are depleted
- ✓ **Reduced mucus secretions** and infection of lungs and urinary tract.
- ✓ **Keratinization of the skin:** The skin is rough, scaly & dry
- ✓ **Delayed growth of bone & teeth.**



Vitamin E

Vitamin E is known also as tocopherol

Sources

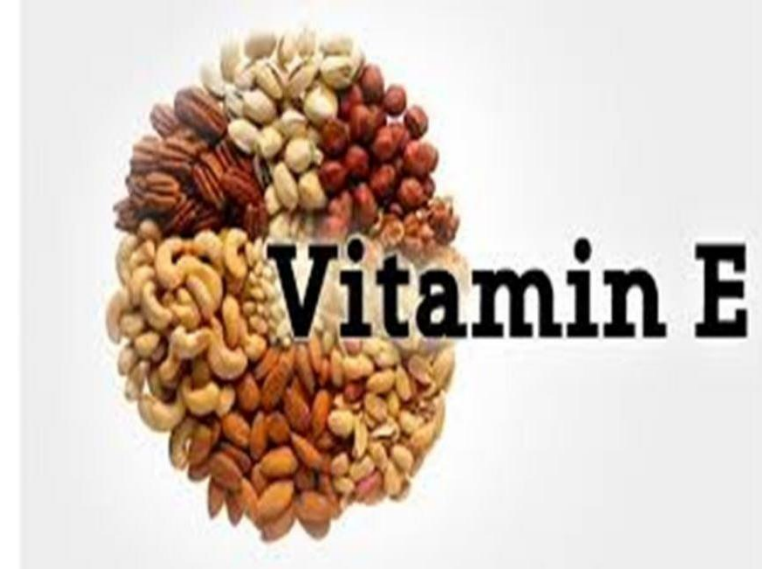
- Plant oils: e.g. Wheat germ, cotton seed & maize oils
- Green leafy vegetables: Lettuce & spinach
- Animal sources: Liver, egg yolk & milk

Functions: Vitamin E is:

- ✓ Natural antioxidant: It protects polyunsaturated FAs (PUFAs) from oxidation & peroxide formation.

Deficiency Manifestations:

- ✓ Hemolytic anemia in children: Due to altered red cell membrane stability leading to shortened life span of RBCs
- ✓ Sterility: Occurs in male rat.



Vitamin K

Sources:

- ❑ **Vitamin K1 (Phylloquinone)**: is natural vitamin K found in plants e.g. green leafy vegetables, cabbage, tomatoes and cereals.
- ❑ **Vitamin K2 (Menaquinone)** : is produced by **intestinal bacteria** and found naturally in animal sources e.g. liver and egg yolk
- ❑ **Vitamin K3 (Menadione)** : is synthetic vitamin.



Functions of vitamin K:

- ❖ Biosynthesis of clotting factors including prothrombin (II) and factors VII, IX and X (2,7,9,10)
- ❖ Is an important antidote for poisoning by anticoagulants (dicumarol like drugs).

Deficiency Manifestations of vitamin K:

- Uncontrolled **bleeding** after a minor trauma.

Deficiency of vitamin K:

Why Deficiency of (vit. K) is rare in adults?

1. Vit. K is widely distributed in plant and animal tissues
2. Vit. K is produced by intestinal bacteria.

Why (vit. K) deficiency is common in newborns?

1. The placenta doesn't pass the vitamin to the fetus efficiently
2. The gut of the newborn is sterile at birth

Diagnosis of vitamin K deficiency:

By measurement of prothrombin time (PT)

- ✓ The normal PT = 16 – 20 seconds
- ✓ Prolonged PT occurs in cases of **vit. K deficiency** or **liver failure** (the liver is unable to synthesize clotting factors).
- ✓ Vit. K administration relieves cases of vit. K deficiency but not in liver failure.

Vitamin C

Vit. C is derived from **glucose** in many mammals

Two (2) forms of the vitamin C are present:

L-ascorbic acid (90%) and **dehydroascorbic acid (10%)**



Sources:

Plant source:

Fresh vegetables and Citrus fruits (good source)

Animal source: Present in little amounts e.g. liver.

Functions:

- ❖ Synthesis of collagen
- ❖ Enhances iron absorption: by keeping iron in the ferrous state (Fe^{+2}).

Deficiency of vit. C produces **Scurvy**.

Manifestations of Scurvy:

- Bleeding tendency: Subcutaneous and other due to weak capillaries defect in collagen synthesis
- Delayed healing of wounds (poor collagen synthesis)
- loosening and falling of teeth & gums: Bleeding, swollen and infected.



Vitamin B complex

Sources

Plant sources: Whole grain cereals and green vegetables

Animal sources: liver , meat, fish, milk and eggs

1. Vitamin B1, thiamin
2. Vitamin B2, riboflavin
3. Vitamin B3, Nicotinic acid or niacin
4. Vitamin B5, pantothenic acid
5. Vitamin B6, pyridoxine
6. Vitamin B7, biotin
7. Vitamin B12, cobalamins
8. Folic acid

Vitamin (B1)Thiamine (Anti Beri Beri vitamin)

Functions

Thiamine pyrophosphate (TPP) is the coenzyme

Deficiency

Its deficiency produces Beri Beri: neurological manifestations as peripheral neuritis, poor memory, muscular weakness and mental confusion + Cardiovascular manifestations lead to oedema.

Vitamin (B2) Riboflavin

(Vitamin B2)

Functions

- FMN and FAD are the coenzymes (for oxidoreductases) act as hydrogen carriers in oxidation reduction reactions.

Vit B3, Niacin or nicotinic acid
(Pellagra preventing factor)



Functions

NAD and NADP act as coenzymes (for oxidoreductases)

Deficiency disease

(Pellagra) pell = rough, agra = skin
characterized by diarrhea.

Called (3d) manifestations

diarrhea,

dermatitis,

dementia

(inflammation of GIT mucosa)

(inflammation of skin, rough scaly)

Vitamin B5, Pantothenic acid

Function:

- ✓ Coenzyme A (COASH)
- ✓ Acyl carrier protein (ACP) for lipid metabolism

Vitamin B6 (Pyridoxine)

Function: Coenzyme pyridoxal phosphate (PLP)

Vitamin B7 (Biotin)

Function: Biotin serves as a coenzyme for carboxylation reactions

Vitamin B12 (Cobalamin).

- Its absorption needs the intrinsic factor which is a glycoprotein secreted by the parietal cells of gastric mucosa.

Function:

Coenzymes: Deoxy adenosylcobalamin,
Methylcobalamin

Deficiency Manifestations

- ✓ Megaloblastic macrocytic anemia accumulation of megaloblasts in bone marrow due to impaired cell division.
- ✓ **Pernicious anemia if the cause is lack of intrinsic factor.**
Neurological manifestations, peripheral neuritis

Folic acid

Function: The coenzyme is tetrahydrofolate (FH₄)

Deficiency:

It is one of the commonest vit. deficiency states in man and common in: during pregnancy. Folic acid when given in pregnancy may prevent formation of neural tube defects, such as spina bifida. About 70% of cases can be prevented by taking folic acid prophylactically.

- ✓ **Pancytopenia** (anemia + leucopenia + thrombocytopenia)
- ✓ **Megaloblastic (macrocytic) anemia** Folic acid can partially reverse the hematologic abnormalities of B12 deficiency and, therefore, can mask a cobalamin deficiency. Thus, therapy of megaloblastic anemia is often initiated with folic acid and vitamin B12 until the cause of the anemia can be determined.



Vitamin	Active Form(s)	Main Functions	Deficiency
 Vitamin A	Retinal, Retinol, Retinoic acid	Vision, epithelial integrity	Night blindness, xerophthalmia
 Vitamin D	Calcitriol	Calcium metabolism	Rickets, osteomalacia
 Vitamin E	α-tocopherol	Antioxidant	Hemolytic anemia
 Vitamin K	Phylloquinone, Menaquinone	Coagulation factors	Bleeding
 B1	TPP	Carbohydrate metabolism	Beriberi, Wernicke-Korsakoff
B2	FAD, FMN	Redox reactions	Cheilosis, glossitis

Vitamin	Active Form(s)	Main Functions	Deficiency
B3	NAD⁺, NADP⁺	Energy metabolism	Pellagra
? B5	CoA	FA synthesis	Dermatitis, neuropathy
? B6	PLP	Amino acid metabolism	Neuropathy, seizures
B7	Biocytin	Carboxylation	Dermatitis, alopecia
? B9	THF	DNA synthesis	Megaloblastic anemia
? B12	Methylcobalamin	Neurologic function	Pernicious anemia
* Vitamin C	Ascorbate	Collagen synthesis	Scurvy



**Thank
You!**

Macrominerals (Major Minerals)

Mineral	Function	Deficiency	Toxicity
Calcium	Bone/teeth, muscle contraction	Osteoporosis, tetany	Kidney stones, hypercalcemia
☐ Sodium	Fluid balance, nerve impulses	Hyponatremia: confusion	Hypernatremia: dehydration
♥☐ Potassium	Heart rhythm, nerve function	Hypokalemia: weakness	Hyperkalemia: arrhythmias
Magnesium	Enzymes, ATP, nerves	Cramps, seizures	Hypotension, bradycardia
Chloride	Acid-base balance, HCl	Metabolic alkalosis	Acidosis, dehydration
☐ Phosphorus	ATP, bone/teeth	Weakness, bone pain	Calcification
☐ Sulfur	Proteins, detox	Rare deficiency	GI discomfort

Mineral	Function	Deficiency	Toxicity
? Iron	Hemoglobin, O ₂ transport	Microcytic anemia	Hemochromatosis
? Zinc	Immunity, healing	Dermatitis, hair loss	Copper deficiency
? Copper	Iron metabolism	Anemia, neutropenia	Liver damage
? Iodine	Thyroid hormones	Goiter, hypothyroid	Thyroid inflammation
Selenium	Antioxidant, thyroid	Keshan disease	Hair/nail loss
? Manganese	Bone enzymes	Poor growth	Neurologic toxicity
? Fluoride	Teeth enamel	Dental caries	Fluorosis
* Chromium	Insulin action	Glucose intolerance	Kidney issues
? Molybdenum	Enzyme cofactor	Rare	Gout-like symptoms
Cobalt	Part of B12	B12 deficiency signs	Polycythemia

Minerals

I. Principal or Macro-minerals:

These are seven elements, which are required in amounts greater than 100 mg / day. The principal elements include: calcium, magnesium, phosphorus, sodium, potassium, chlorine and sulfur.

II. Micro-minerals (trace elements):

These are required in amounts less than 100 mg /day. They are 15 elements, include: iron, copper, zinc, manganese, iodine, cobalt, molybdenum, selenium and chromium. >

1- Macrominerals

Calcium

» Sources:

- Dairy products: milk, cheese and yoghurt.
- Beans and leafy vegetables.

Blood calcium:

- » About 99% of the body Ca^{++} is present in the skeleton (bones and teeth). The remaining 1% is present in other tissues and body fluids.
- » The plasma contains 9-11 mg 100 ml
- » The erythrocytes contain no calcium.
- » 45% of plasma calcium is bound to plasma proteins mainly albumin and is physiologically inactive. This is the non diffusible non ionized form. >

- » 5% inactive diffusible non ionizable calcium citrate.
- » The only physiologically active form of calcium is the ionizable diffusible form which constitutes 50% of plasma calcium.
- » If the ionizable form decreases, tetany occurs.
- » Functions of calcium:
 - » Calcium is a constituent of bone and teeth.
 - » It is important in the regulation of nerve impulses and muscle contraction.
 - » Calcium is essential for blood and milk clotting.
 - » It is essential for the activity of some enzymes e.g. lipase and phosphorylase.
 - » Calcium is important for the regulation and secretion of some hormones e.g. insulin.

Phosphorus

Sources:

- Milk, cheese and yoghurt.
- Liver, kidneys, meat and fish are good sources.

Functions:

- Phosphorus enters in the formation of bone and teeth
- It enters in the formation of sugar phosphates, phospholipids, phosphoproteins nucleic acid, nucleotides and many coenzymes e.g. TPP, NAD, NADP, FAD, FMN, CoASH and creatine phosphate.
- It is a part of blood buffer.

Deficiency disease:

Osteomalacia in adults & rickets in children.



Sodium

Sources:

- Table salt (Salt added to prepared food).

Blood sodium level:

Plasma : 135-150 mEq/L.

The metabolism of sodium inside the body is regulated by the hormone aldosterone.



Functions of sodium:

1- Maintain osmotic pressure and the extracellular fluid and plasma volumes.

2- It is the principal cation of the extracellular buffers.

3- Sodium regulates acid-base balance.

4- It is important for transmission of nerve impulses.

5- Sodium is important for contraction of muscles.

Hyponatraemia leads to muscular cramps.



Potassium

Sources:

Animal sources liver, kidney, heart, meats and fish.

Plant sources: vegetables, fruits and nuts.

Blood potassium levels:

Plasma: 3.5-5 mEq/L.

If the plasma levels of potassium are increased, this leads to a condition called **hyperkalaemia**. Whereas if the plasma levels are decreased, this leads to **hypokalemia**.



Functions:

- » Potassium is the principal cation in intracellular buffers, it maintain osmotic pressure and intracellular fluid volume.
- » Important for transmission of nerve impulses.
- » Important for contraction of muscles.
- » Essential for the activity of Na/K ATPase.



Chloride

Sources: table salt.

Blood level:

Plasma: 96-106 mEq/L.

Functions

- » Chloride is the main extracellular anion, essential for fluid and electrolyte balance.
- » It is essential for formation of hydrochloric acid in the gastric fluid.
- » Activation of enzymes as salivary and pancreatic amylase.



Magnesium

Functions:

- » Magnesium is an important constituent of bones and teeth.
- » It acts as an enzyme cofactor e.g. kinases.
- » It is important for normal muscle contraction, transmission of nerve impulses and it decrease neuromuscular excitability.



Microminerals (Trace elements)

Iron

» Sources:

- Red meat, liver and egg yolk are very good sources.
- Molasses, dates, legumes, vegetables and cereals.

» Distribution & function:

The total body iron is 4g, it is present in two forms:-

- ☐ Hem containing compounds
- ☐ Non-Hem containing compounds



» Heam-containing compounds(80%)

1- Hemoglobin (70%): This is the main form of iron in the body. Hemoglobin carries oxygen from the lungs to the tissues and CO₂ in the opposite direction.

2- Myoglobin (5%): It is found in muscles. Myoglobin functions as temporary carrier for oxygen that releases it during muscular contraction.

3- Respiratory enzymes (5%)



B) Non-haem containing compounds (20%)

These include transport and storage forms of iron.

1- **Transferrin:** is the transport form of iron in the blood.

2- **Ferritin:** is the storage form of iron in the tissues.

3- **Haemosiderin:** This occurs only when the body contains excess iron.

4- Some **flavoprotein enzymes:** These include NADH dehydrogenase and succinate dehydrogenase.



» **Toxicity symptoms:**

Excessive administration or absorption of iron leads to hemosiderosis. There is deposition of iron in the tissues including skin, liver and pancreas (bronzed diabetes).

» **Deficiency:**

Iron deficiency occurs secondary to malnutrition and leads to hypochromic microcytic anemia.





**Thank
You!**